Lecture 11: Early CAPM Tests and First Cracks

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Intro

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- We now start the factor-model module in empirical asset pricing
- Today is about the first CAPM tests and the first cracks in the model
- Logic: CAPM benchmark first, data second, anomalies third, joint tests last

Flight Plan

- Part I: why CAPM looked compelling in theory (Sharpe 1964; Lintner 1965; Mossin 1966)
- Part II: early empirical protocols from Black, Jensen, and Scholes (1972) and Fama and MacBeth (1973)
- Part III: the first anomaly evidence (Basu 1977; Banz 1981; De Bondt and Thaler 1985)
- Part IV: joint testing logic, from Roll (1977) to Gibbons, Ross, and Shanken (1989)

Part I: CAPM Recall

Early Days

- In 1952: Markowitz brings us to light with “Modern Portfolio Theory’ ’
- Key insight: you should diversify your investments, correlation matters
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In the 1960s:

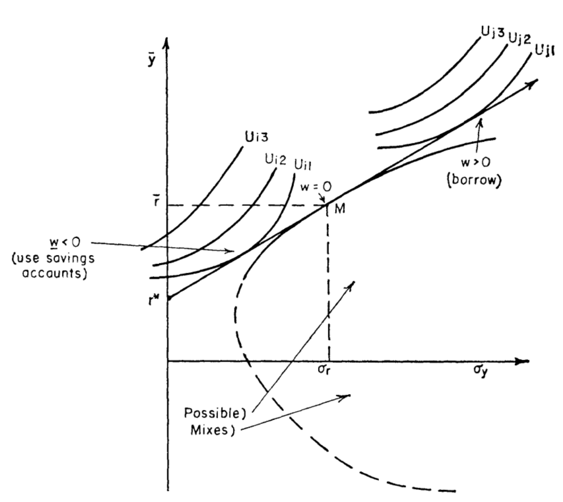
- Sharpe (1964), Lintner (1965), and Mossin (1966) build on Markowitz to get the CAPM
- Notice this *way* before the Rational Expectation revolution and the SDF framework
- These guys were The Beatles of Finance!

What CAPM Delivered

- It linked portfolio choice to equilibrium expected returns
- It delivered one sufficient statistic: market beta
- Investors are mean-variance optimizers over one period: you like returns, dislike variance
- Homogeneous beliefs and frictionless trading are assumed
- Everyone can borrow/lend at one risk-free rate and hold the same assets
- Important: different investors will have different preferences!

Mean-Variance Frontier and Tangency Portfolio

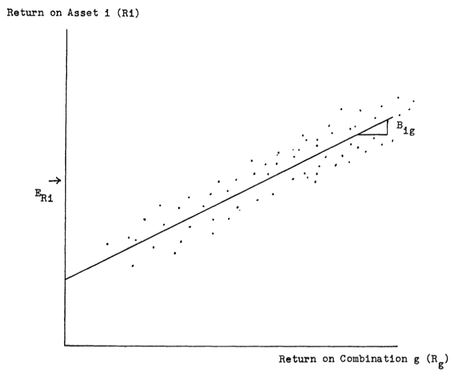
- Efficient portfolios are combinations of the risk-free asset and *one* tangency portfolio
- Tangency portfolio: highest Sharpe-ratio risky portfolio
- Investors differ only in how much risk they scale, not in risky composition
- They will *not* have the same overall portfolio!



Security Market Line

- Expected excess returns are linear in market beta
- Only systematic market covariance is priced
- Idiosyncratic risk earns no premium

$$\mathbb{E}[R_i - R_f] = \beta_i \, \mathbb{E}[R_m - R_f], \quad \beta_i \equiv \frac{\operatorname{cov}(R_i, R_m)}{\operatorname{var}(R_m)}$$



CAPM as SDF/Beta Representation (Bridge to Lecture 4)

$$m_{t+1} = a - bR_{m,t+1}$$

- CAPM is one special case of the general SDF framework from Lecture 4
- If the SDF is affine in market returns, beta pricing follows mechanically
- This gives a clean bridge between no-arbitrage language and equilibrium CAPM
- The idea of Bansal and Yaron (2004) to think about a “wealth portfolio” is much older than you think!

Testable Implications for Data

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i (R_{m,t} - R_{f,t}) + \varepsilon_{i,t}$$

- Cross-section: higher beta assets should have higher average returns
- The relation should be approximately linear
- Exact CAPM implies $\alpha_i = 0$ for all test assets
- Nonzero alpha patterns indicate systematic model failure
- Important: nothing else should be added to the right-hand side!

Questions?

Black (1972): Zero-Beta CAPM

$$\mathbb{E}[R_i] = \mathbb{E}[R_z] + \beta_i(\mathbb{E}[R_m] - \mathbb{E}[R_z])$$

- Black (1972) relaxes unrestricted risk-free borrowing and lending
- A zero-beta portfolio replaces R_f as the intercept object
- The linear beta-pricing equation survives with a different baseline return

Zero-Beta CAPM: Derivation Intuition

$$\mathrm{cov}(R_z, R_m) = 0, \qquad \beta_i \equiv \frac{\mathrm{cov}(R_i, R_m)}{\mathrm{var}(R_m)}$$

- Without risk-free borrowing/lending, the efficient set is generated by risky portfolios only
- The relevant baseline return is the expected return on the portfolio orthogonal to R_m
- Orthogonality gives the same slope object (β_i), but a different intercept
- Intercept shifts from R_f to $\mathbb{E}[R_z]$

Why Zero-Beta Matters for Empirical Work

- Data often reject the strict Sharpe-Lintner intercept restriction
- A nontrivial zero-beta return can absorb part of that mismatch
- This reframed tests as “is linear beta pricing there?” rather than “is intercept exactly R_f?”
- It directly shaped how Black, Jensen, and Scholes (1972) interpreted evidence

Questions so far?

Part II: Early CAPM Tests in Data

From Theory to Data: What Early Tests Actually Did

- Build test assets with different estimated betas
- Estimate time-series loadings and pricing errors
- Run cross-sectional regressions linking average returns to betas
- Evaluate whether CAPM restrictions hold jointly

Black-Jensen-Scholes (1972): Design + Regression Setup

- Core idea: sort assets by historical beta to get spread in risk exposure (Black, Jensen, and Scholes 1972)
- Use portfolio returns to reduce idiosyncratic noise
- Run the standard time-series test in each beta-sorted portfolio:

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_p(R_{m,t} - R_{f,t}) + \varepsilon_{p,t}$$

- Compare average returns across low- and high-beta portfolios
- Check whether estimated intercept behavior matches strict or zero-beta CAPM

BJS Main Findings + Legacy

- Beta and average return are positively related, but flatter than strict CAPM predicts
- Low-beta portfolios earn “too much” relative to strict CAPM
- High-beta portfolios earn “too little” relative to strict CAPM
- Evidence is more consistent with a zero-beta intercept than with the risk-free intercept
- Set the template for beta-sorted portfolio tests and alpha diagnostics

BJS Evidence: SML Flattening (Figure Placeholder)

[Insert BJS SML flattening plot/table here]

BJS Evidence (Table/Figure Placeholder)

[Insert BJS beta-portfolio table/figure here]

Fama-MacBeth (1973): Why Two Passes?

- Exposure estimation and risk-price estimation are different statistical problems
- Two-pass procedure separates those two steps cleanly (Fama and MacBeth 1973)
- It produces a time series of risk-price estimates for inference
- This became the default empirical workflow in cross-sectional asset pricing

Fama-MacBeth Pass 1 (Estimate Betas)

$$R_{i,t} - R_{f,t} = a_i + b_i(R_{m,t} - R_{f,t}) + e_{i,t}$$

- First-pass time-series regressions estimate each asset's market beta
- Betas are treated as characteristics for the second-pass cross section
- Portfolio grouping can be used to stabilize estimates

Fama-MacBeth Pass 2 (Price of Risk)

$$R_{i,t} = \gamma_{0,t} + \gamma_{1,t}\hat{\beta}_i + u_{i,t}$$

- At each date, run a cross-sectional regression of returns on estimated betas
- Average $\gamma_{1,t}$ over time to estimate the market price of beta risk
- Average $\gamma_{0,t}$ to assess the intercept restriction

Fama-MacBeth Main Findings + Limits

- A positive average beta-return relation is present
- Intercept evidence is not cleanly aligned with strict Sharpe-Lintner CAPM
- Still the backbone workflow, but with generated-regressor and beta-estimation caveats
- Conclusions depend on test assets, sample period, and market proxy quality

Fama-MacBeth Risk Price Path (Table/Figure Placeholder)

[Insert time-series of $\gamma_{1,t}$ */ rolling estimates here]*

Fama-MacBeth Evidence (Table Placeholder)

[Insert Fama-MacBeth second-pass estimates table here]

If beta is enough, why did anomalies show up so quickly?

Part III: The First Cracks

Basu (1977): Characteristics vs Beta

- Basu (1977) sorts stocks by earnings-to-price ratios
- High E/P portfolios earn higher average returns than low E/P portfolios
- This spread remains difficult to map into a pure one-beta CAPM story
- Early signal that characteristics may contain pricing information
- CAPM says beta should span expected-return differences; Basu says E/P also forecasts returns

Banz (1981): Size Effect

- Banz (1981) documents higher average returns for small-cap stocks
- The size-return spread is hard to eliminate with market beta alone
- Small-minus-big logic later becomes a canonical factor construction
- Another direct empirical crack in single-factor CAPM

DeBondt-Thaler (1985): Long-Run Reversal

- De Bondt and Thaler (1985) show that long-horizon losers tend to outperform winners
- Multi-year return reversals are difficult for static CAPM to rationalize
- Short takeaway for this lecture: return dynamics matter for pricing tests
- Behavioral interpretations enter the CAPM debate more directly

What These Cracks Have in Common

- They generate systematic nonzero pricing errors relative to CAPM
- Firm characteristics and past returns carry incremental information
- One-factor beta seems too narrow for the observed cross section
- The field now needs stronger model-comparison tools

Early Anomalies (Figure/Table Placeholder)

[Insert anomaly spreads figure/table here]

Part IV: Joint Testing and Synthesis

Why We Need Joint Tests (Roll-to-GRS Motivation)

- Testing each asset separately creates a multiple-testing problem
- A valid model should price a set of assets jointly, not one by one
- Roll (1977): every CAPM test is joint with the chosen market proxy
- The natural next step is a formal joint test on the alpha vector

GRS Setup and Null

$$R_t^e = \alpha + Bf_t + \varepsilon_t, \qquad t = 1, \dots, T$$

$$H_0 : \; \alpha = 0_N$$

- Gibbons, Ross, and Shanken (1989) propose an exact finite-sample test of this joint null
- Rejection means the tested factor portfolio is not mean-variance efficient for that asset set
- Notation: N test assets, K factors, T time observations

GRS Test Statistic

$$GRS = \frac{T-N-K}{N}\left(1 + \hat{\mu}_f' \hat{\Sigma}_f^{-1} \hat{\mu}_f\right)^{-1} \hat{\alpha}' \hat{\Sigma}_\varepsilon^{-1} \hat{\alpha} \sim F_{N,\,T-N-K} \text{ under } H_0$$

- $\hat{\alpha}$: vector of estimated pricing errors from the time-series regressions
- $\hat{\Sigma}_\varepsilon$: residual covariance matrix across test assets
- $\hat{\mu}_f, \hat{\Sigma}_f$: sample mean and covariance of factor returns

GRS Practical Insight + Figure Space

- Good models need jointly small alphas, not only scattered insignificant t-stats
- Statistical rejection and economic materiality are different objects
- Conclusions depend on which assets and factors are included in the test
- GRS remains a core diagnostic for factor-model performance

[Insert GRS p-value / alpha heatmap figure here]

Taking Stock

- CAPM gave a disciplined starting benchmark
- BJS and Fama-MacBeth established the empirical playbook
- Basu, Banz, and DeBondt-Thaler revealed systematic cracks
- GRS formalized joint model evaluation in finite samples
- Next step: the Fama-French revolution and the multi-factor era

Readings and References i

- Foundational CAPM papers: Sharpe (1964); Lintner (1965); Mossin (1966)
- Zero-beta and early tests: Black (1972); Black, Jensen, and Scholes (1972)
- Two-pass cross-sectional method: Fama and MacBeth (1973)
- Early cracks: Basu (1977); Banz (1981); De Bondt and Thaler (1985)
- Joint-testing framework: Roll (1977); Gibbons, Ross, and Shanken (1989)

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